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Chapter 1

Maintenance



Maintaining Configurator

The maintenance options are presented here in the order in which you need to use them to set up Configurator. The company profile option is included because it is in effect a maintenance-style option.



Tip: The item maintenance options from Inventory and Production are all available from a single option, Linked Item Maintenance. Use this option when setting up your items to avoid having to switch between menus and applications.

Pricing the Configured Item

The pricing of your configured items is dependent on the setting of the pricing method in the company profile. There are three settings:

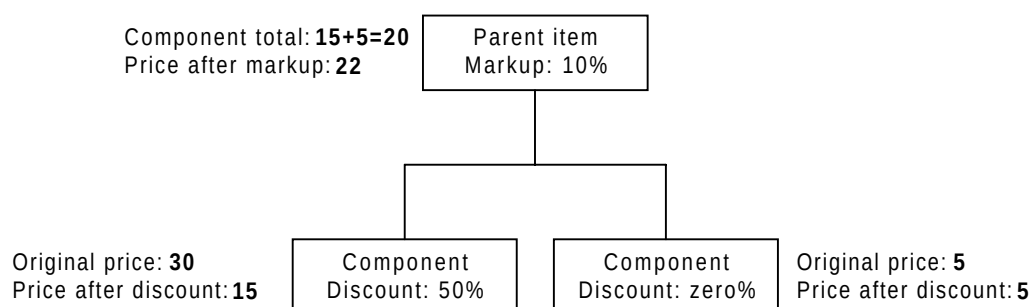
- Pricing method=0, where the price is taken directly from the price list for the parent item.
- Pricing method=1, where the price is the total of all the component prices, taken from component price lists.
- Pricing method=2, where Configurator checks the Inventory item cost, and takes the cost stored in the Configurator cost list, for each component. This is added to the operation costs, picked up from the Production standard cost set. A margin is then added to this total using a discount list.

Price lists for methods 0 and 1 are set up as explained in the Sales Order Processing product information, and work according to standard SOP procedures.

Pricing Method 2 - Details

To set up Pricing Method 2 do the following:

1. Define operation costs on the Production route.
2. Run the Production Single Item Recost option to write these Production costs to a file accessed by Configurator. Create a cost price list profile for components.
3. Enter 1 in the Configurator 'cost list' field to indicate that the list holds costs (if the profile is a price list) or discounts and margins (if the profile is a discount list).
4. Run the Component Costing option to write costs to the cost price lists.
5. Set up the cost discounts for components, if discounts are to be applied, and in the same list set up the mark-up percentage for the parent item. This is done by entering a negative value.



When both cost and margin profiles have been set up, go to the Component Costing option (see the next section), which automatically writes the existing component costs to the lists.

You then enter the discount list option, enter the configured item, and set up the margin which you want to be added to the cost price. This must be entered as a negative value.

Advanced Pricing

If advanced pricing is active for the company (this is specified on the SOP company profile), configured lines will be priced as follows:

Pricing method 0

The configured item is priced using the Advanced Pricing price and discount profiles, in exactly the same way as for normal order lines.

Pricing method 1

The individual components for a configuration are priced using the Advanced Pricing price and discount profiles. The component prices are rolled up to give the total price for the configured item. The total price is registered as a fixed price for the sales order line.



Note: Advanced Pricing will always be performed interactively for configured lines if this pricing method is used.

Pricing method 2

This method does not use Advanced Pricing function. You can use Advanced Pricing for normal order lines, but configured lines will be priced using standard pricing function.

Search Family / Characteristic

In Advanced Order Entry, you use this function to define codes and characteristics for search family and Certificate of Conformance groups. For details of how search groups and Certificates of Conformance are used, refer to the Advanced Order Entry product information.

However, within Configurator, search family codes have more specific uses:

- To link a model item to related base items.
- To link generic components to related actual components.
- To define individual characteristic codes and values to aid the search and selection of base items and components during configuration. (To search for a matching component based on the specified characteristics)
- To link a feature with its optional components.



The search family is set up in three stages.

Stage 1

Create a description file major type and associated code for each of the characteristics to be defined to the family, for example:

Characteristic Type:	Paint colour
Description File Major Type:	'COLR'
Description codes:	'RED BLUE GREEN'

Stage 2

Define the search family itself:

Define new search family	PNT (Paint)
Define up to 10 characteristics to the search family:	FNSH (Finish) COLR (Colour)
Select each characteristic from the description file to define the ones allowed within this search:	Matt - Red Matt - Blue Gloss - Blue

Stage 3

Within Inventory Item Maintenance, define the search family to which the component belongs together with its characteristics.

Component item:	PBLMT (Blue matt paint)
Search Family Code:	PNT (Paint)
Characteristic 1:	COLR=BL (Blue)

Characteristic 2: FNSH=MT (Matt)

This is done for each component item within the family.

Search family codes and their related characteristic values are defined to individual items in Inventory using the Items option.

The search family contains up to 10 characteristics and each characteristic can have an unlimited number of defined values.

The components assigned to the search family each have a set of characteristic values.

Example:

Generic component	Paint
Search family	PNT (Paint)
Actual Component #1	PBLMT - (Blue matt)
Characteristics of #1	Paint - 'Blue'
	Finish - 'Matt'
Actual Component #2	PRDGL - (Red gloss)
Characteristics of #2	Paint - 'Red'
	Finish - 'Gloss'



You cannot change the decimal place details once they have been entered.

This field must be blank if the characteristic is associated with a characteristic code.

Characteristic Code

You can optionally define each characteristic to a characteristic code. If so, the code must have been set up as a major type in Inventory Descriptions. This facility allows a set of permitted values to be defined to the characteristic.

Characteristic Code Description

The text defined against the characteristic code in Inventory Descriptions.

Number Of Values

The number of different values currently allowed for the characteristic code. Enter 1 against a characteristic code to maintain it.



When the required characteristics have been entered, press **F8** to update the details.

Functions:

F5

Refresh the screen to display data as at last update.

F8

Update the details and return to the invite screen.

F11

Delete the search family code and its associated characteristics. Deletion is not allowed if the code is already attached to an item as a search family.

Characteristic Values Window

 Enter 1 in the Select field on the Search Value / C of C Code Maintenance Screen to display the Characteristic Values window.

You use this window to select a set of values for the characteristic. The list displayed is the list of values set up for this characteristic using the Inventory Descriptions option. This means that you can set up different sets of permitted values for the same characteristic code in different search family codes.

IN200

Z1 - Demonstration Company

User V3USER

17/10/95
15:37:01

Search Family / Certificate of Conformance Code Maintenance

Code..... PNT

Description..... Pain

	Present	Characteristics
Sel	Seq	Description
1	01	Colour
-	02	Finish
-	-	-
-	-	-
-	-	-
-	-	-
-	-	-
-	-	-
-	-	-
-	-	-

Characteristic Values

Seq	Code	Description
1	BL	Blue
4	CL	Clear
2	GR	Green
3	RD	Red

F8=Update F12=Previous

1=Select

F3=Exit F4=Prompt F5=Refresh F8=Update F11=Delete F12=Previous

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Fields:

Characteristic Code

The characteristic code for which permitted values are being maintained.

Code

Lists the actual values for the characteristic.

Description

Text describing the characteristic value as defined in the Inventory Descriptions option.

Sequence

Select values by specifying the sequence in which they are to appear on displayed lists.

Company Profile

This option controls the workings of Configurator, and you should carefully consider all settings for their far-reaching implications.



Warning: Extreme care must be taken when you amend existing profiles because of the effect this may have on data already in the application.

Configurator Company Profile Maintenance Screen 1



Select Company Profile Maintenance to display the first Configurator Company Profile Maintenance screen.

You use the Configurator Company Profile Screen to maintain the Company Details.

CN000	Z1 - Demonstration Company	User V3USER	17/10/95
			15:39:33

Configurator Company Profile Maintenance

Configuration Route..... 0

Item Creation..... 0 (0=Batch, 1=Interactive)

Item Code Generation..... 1 (0=Manual, 1=Automatic)

Base Item Code Length.... 4

Item Code Delimiter..... /

Variation Number Length. 4

Item Categories

Model Items.....?.. M

Base Items.....?.. B

Preconfigured Items...?.. P

Order Configured Items?.. 0

Generic Components....?.. G

Update Audit Files..... 0 (0=No, 1=Yes)

F3=Exit F4=Prompt

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Fields:

Configuration Route

Specify the Production route code which will be used as the configuration route for this company. This is the only route which can be used for configuration for this company.

The configuration route is used when defining routes for model and configured items.

Model items and order configured items must be defined to this route code in route maintenance. Newly defined preconfigured items will be assigned this route code at the end of preconfiguration.

Item Creation

Enter one of the following:

0 - to create preconfigured item details in batch.

1 - to create the details interactively.

Preconfigured item details are created at the end of preconfiguration, based on the modified model route, and the item details set up for the base item.

Item Code Generation

Enter one of the following:

0 - to manually enter item codes during preconfiguration.

1 - to allow the system to generate codes automatically. Generated codes are constructed from a base item code prefix and are suffixed by a unique sequence number.

Specify how item codes for preconfigured items are to be set up at the end of preconfiguration.

Base Item Code Length

If preconfigured item codes are to be automatically generated, specify the number of characters from the base item to use as a prefix.

Item Code Delimiter

If preconfigured item codes are to be automatically generated, specify the character to appear between the base item code prefix and the unique sequence number.

Variation Number Length

If preconfigured item codes are to be automatically generated, specify the number of figures to appear in the unique sequence number suffix.

For example, a variation number of 3 will allow up to 999 unique preconfigured item codes to be generated. A variation number of 5 will allow 99999 unique codes.

Item Categories

Specify the Inventory source flag value that is to identify each kind of item.

Each value must have been set up via the Inventory Descriptions option, major type 'INSC', and each should be reserved for the definition of its particular kind of item (model, base etc.) in the Inventory Items option.

Model items are used to define the high-level route which contains all possible operations and actual or generic components for a particular group of configurable products. The model route is then used as the basis for the definition of route details of individual configured items in preconfiguration.

Base items are used to define the item master records from which individual configured item details are copied during preconfiguration. It is the entry of a model / base item combination in Route Maintenance which forms the basis for the definition of a new preconfigured item. Base items may also be used to attach parent-type rules to a configuration, or to aid the search and selection of a preconfigured item during order entry.

Preconfigured items are generated in preconfiguration from a model / base item combination. When an item master record is written for the new preconfigured item it will automatically be updated with the Inventory source value entered here.

Order configured items are created in Inventory, then have a route created for them in Production. No model or base item is required, and the preconfiguration process is unnecessary. Configurator ensures that all features and options are presented during order entry.

Generic components define the 'features' of a configuration. When these generic components are defined to the configuration route, they are replaced during the process of configuration by actual components selected from the same search family.

Update Audit Files

Enter one of the following:

0 - do not create audit records.

1 - create audit records whenever rules and messages are maintained.

Auditing requires disk space, so use it only if you know you have a requirement for the information.

The following are the Configurator audit files:

CNP40A – Rule header audit

CNP42A – Rule detail audit

CNP44A – Route rules audit

CNP46A – Configurator messages audit



Press **Enter** to see the second Configurator
Company Profile Maintenance screen.

Configurator Company Profile Maintenance Screen 2



Press **Enter** on the first Configurator Company Profile Maintenance screen to display the second screen.

You use this screen to maintain the Configurator Company Profile.

CN000	Z1 - Demonstration Company	User GBJBANB1	27/10/95 9:59:22
Configurator Company Profile Maintenance			
Order Entry Costing.....	E (0=No, 1=Operations, 2=Components, 3=Both)		
Costing of Despatches....	1 (0=Despatch Cost, 1=Order Entry Cost)		
Pricing Method.....	2 (0=Configured Item, 1=Component Roll-Up, 2=Cost Plus Markup)		
Selection Method.....	1 (0=Direct, 1=Prompted, 2=Unprompted)		
Order Analysis Level.....	2 (1=Operations & Components, 2=Characteristic)		
F3=Exit F12=Previous			
05-35	SA	MA	KS IM II KENB20A PET149S1

Fields:

Order Entry Costing

This field is relevant only if the pricing method is 2.

- 0 - No costs
- 1 - Roll up operation costs only
- 2 - Roll up component costs only
- 3 - Roll up component and operation costs

Costing Of Despatches

- 0 - Inventory costs at despatch. This can be used with any pricing method.
- 1 - Order entry costs. This is a valid option only if the pricing method is 2.

Pricing Method

- 0 - Price / discount lists set up for each configured item
- 1 - Roll up prices of all components from price list
- 2 - Roll up cost of all components and / or operations, plus mark-up percentage defined for the configured item on a discount list

If you choose option 2 then the Order Entry Costing field must hold one of the rollup options (1, 2 or 3).

This field determines how your configured item will be costed.

Selection Method

This field determines whether you are prompted with a window to help you select the values you want. There are three possibilities, direct, prompted and unprompted.

Different user IDs can be set up with individual prompt values, so that more experienced users are not prompted while others have prompts. This is set up by entering the user IDs as description codes under major type 'USID' in the Inventory Descriptions option. The VAT flag is used to indicate the choice (1 - Prompted, 2 - Unprompted, Any other value - Direct).

You can change your chosen prompt at any time, e.g. one simple feature may be selected in direct mode and a more complex one in prompted mode.

0 - Direct - (lists components)

The 'direct' mode is where the components are represented by item codes and descriptions, and you select one from the list:

Item	Description
PBLGL	Paint - blue gloss
PBLMT	Paint - blue matt
PRDGL	Paint - red gloss

This is the fastest way of selecting options, because all variants are displayed on one screen.

1 - Prompted (lists characteristics)

The 'prompted' mode displays each characteristic in a separate window, listing all available options:

Value	Description
GL	Gloss finish
MT	Matt finish

When you have selected your finish, the following colours will be listed for selection:

Value	Description
RD	Red paint
GR	Green paint
BL	Blue paint
CL	Clear varnish

2 - Unprompted (enter characteristics directly)

The 'unprompted' mode shows a list of characteristics. You must then actively enter the value required. A prompt is always available, listing the values:

Characteristic	Value (input field)
----------------	---------------------

Finish Input as required

Colour Input as required

Order Analysis Level

1 - Operation and component details are recorded for analysis. Phantoms remain unexploded.

2 - Operation, component and characteristics are recorded for analysis.
Phantoms remain unexploded.

This option enables you to store the particular features and options selected for a configured order. The information is kept in the analysis file CNP90, and can then be manipulated as required.

There are storage considerations involved. The amount of information stored by option 2 depends on the number of characteristics defined and used.



Press **Enter** to update and exit.

Configurator Rules

You use this option to set up the rules which are to govern the configuration of any configurable item. These rules specify the inclusion or exclusion of operations, components, characteristics or text formats for the item.

These rules, once created using this option, are specified when you are setting up the route, and can be applied to the configuration during preconfiguration and while a sales order is being entered.

Each rule test can be based on detecting a unique value for any of the following:-

- C - Characteristic
- I - Component item code
- O - Operation number or code
- P - Parent item code (i.e. the base, preconfigured or order configured item code)

The action to be taken if the test is satisfied can be the automatic inclusion or exclusion of any of the components or operations when building the configured product.

A user-defined message can then be issued which may be critical (in which case you cannot proceed without amending the current configuration), or memo (the message can be overridden).

Each rule must be assigned to a rule category. These categories must have been set up in Inventory Descriptions (RCAT) already. You can use the categories to help you search for a particular rule. You can if you wish define and use one category only. The categories must be codes of not more than 5 characters in length.

The rules have a common format:

If rule test is satisfied

then invoke rule actions:

Rule	Comment
If 'MATL=PVC' and O='PNT'	If you have selected the paint operation for a PVC desk...
Then MSG001='You cannot paint a PVC surface'.	...then display this message.
If I 'SHELF=1'	If the optional computer shelf is selected, include
Then INCL O 'SHF'	the 'fix shelf' operation.



Select Configurator Rules, then enter a new or existing rule code. Press **Enter** to see the first Maintain Configurator Rules screen.



Press **F13** to display the second Maintain Configurator Rules screen.

Maintain Configurator Rules Screen 2 (F13)



Select **F13** on the Maintain Configurator Rules Screen 1 to display the second Maintain Configurator Rules screen.

You use this screen to maintain rule tests and rule actions for your selected rule, and, if required, to attach messages to a rule.

The screen is split into two sections. In the upper section, rule tests are entered. In the lower section, corresponding rule actions are entered.



Warning: You must not enter a message as a rule action on its own; an actual action must be specified as well as a message.

For example, instead of specifying the rule test as:

If paint=varnish

And finish=matt

and the rule action as:

Then display the message 'Varnish cannot be matt',

you should specify the test as:

If paint=varnish

and the action as:

Then exclude finish=matt

And display the message 'Varnish cannot be matt'.

CN005		Z1 - Demonstration Company		User V3USER		17/10/95	
						15:43:08	
Maintain Configurator Rules							
Rule Number.....		DSK020		Desk all PVC or all pine			
Category.....		CAT02		Desk component rules			
Test	SFC		Operation	Test	Value (?)	And	
Type	(?)	Item (?)	(?)			/Or	
-	C	SID	MATS	EQ	PVC	AND	
-	C	TOP	MATS	NE	PVC		+
4=Delete							
Actn	SFC		Operation	Test	Value (?)		
Type	(?)	Item (?)	(?)				
-	C				MSG020		+
4=Delete							
F3=Exit F4=Prompt F8=Update F12=Previous F13=Messages							
10-02 S2 MW KS IM II KENB20A PET147S1							

Rule tests may comprise either a single test, multiple 'and' tests or multiple 'or' tests.

Rule actions specify what will be done when all the tests for the rule are satisfied during configuration. Rule actions may consist of the inclusion or exclusion of operations, components, all components with particular characteristics or specific text formats.

Fields:

Delete

Enter 4 to delete a rule test.

Test Type

Specify a code to indicate the type of rule test:

I - To test components.

P - To test parent items (i.e. base or configured item codes).

C - To test item search characteristics.

O - To test operations.

S - To test standard variables (for example, to find out the quantity ordered).

There is a set list of variables you can use in the test, and these are:

- Price of component (base)
- Price of component (prime)
- Unit price of component (base)
- Cost of component (base)
- Total cost of configured item (base)
- Customer
- Customer delivery sequence
- Stockroom
- Quantity ordered
- Quantity per.

V - To test runtime variables. Use this if you need to match a variable. For example, if a customer orders Item 1 with Accessory 4, the system can ensure that Item 1 is the same colour as Accessory 4, without your having to set up an individual rule for each individual colour.

Sfc

For a test on item search characteristics or variables, enter the search family code to which the characteristics belong. Leave blank for all other test types.

Item

For a test on items (component or parent), or for variables, enter the required item code. For a characteristic test, enter a characteristic code within the selected search family group. Leave blank for an operation test.

For a standard variable test, you must enter the relevant code from the list below. Select the variable you want to test on, and enter the corresponding code in the Item field:

- Price of component (base) - *BPRI
- Price of component (prime) - *PRIC
- Unit price of component (base) - *UNPR
- Cost of component (base) - *CCST
- Total cost of configured item (base) - *COST
- Customer - *CUSN
- Customer delivery sequence - *DSEQ
- Stockroom - *PSTR
- Quantity ordered - *QTOR
- Quantity per - *QPER.

Operation

For an operation test, enter the code of the operation. Leave blank for all other test types.

Test

A characteristic test may be set up to check a characteristic value against the value specified in the test according to any of the following conditions:

EQ - Equal to

NE - Not equal to

GT - Greater than

GE - Greater than or equal to

LT - Less than

LE - Less than or equal to

For an operation test, only the conditions EQ (equal to) or NE (not equal to) are allowed.

Leave blank for item test types.

Value

For a characteristic test only, enter the actual value of the specified characteristic that is to be tested for. For a variable test enter the literal *VALUE. Leave blank for all other test types.

And/Or

For multiple rule tests, enter AND on the first test line if all tests must be fulfilled to trigger the action of the rule. Enter OR if only one test needs to be fulfilled.

Delete

Enter 4 to delete a rule action.

Actn

I - Include the item, characteristic, operation or text format specified.

E - Exclude the item, characteristic, operation or text format specified.

C - The message attached to this action is to be 'critical'. This means that the message, once displayed, will show until the error condition is rectified.

M - The message attached to this action is to be 'memo'. This means that the message, once displayed, can be ignored and processing can continue.

Actn Type

Specify a code to indicate the type of rule action:

I - An action on component item codes.

T - An action on text formats.

C - An action on item search characteristics. This is the only action type allowed for a runtime variable test.

O - An action on operation codes.

Sfc

For an action on item search characteristics, enter the search family code to which the characteristics belong. Leave blank for all other action types.

Item

For an action on component items, enter the required item code. For a characteristic action, enter a characteristic code within the selected search family group. Leave blank for an operation or text action.

Operation

For an operation action, enter the code of the operation. Leave blank for all other action types.

Test

A characteristic action may be set up to check a characteristic value against the value specified in the action according to any of the following conditions:

EQ - Equal to

NE - Not equal to

GT - Greater than

GE - Greater than or equal to

LT - Less than

LE - Less than or equal to

For an operation or text action, only the conditions 'EQ' (equal to) or 'NE' (not equal to) are allowed.

Leave blank for item action types.

Value

For a characteristic action only, enter the actual value of the specified characteristic that is to be tested for. For a runtime variable test enter the literal *VALUE. Leave blank for all other action types.

If this line is indicating that a message has been attached to the action, the message code will be displayed here. It is not maintainable.



Note: Attach messages to the rule actions by pressing F13. Once a message is attached, it is shown by a C (critical message) or M (memo message) in the action field. You can change the message severity (C or M) here. The code of the message is displayed in the Value field.



Press **F8** to update rule details.

Functions:

F13

Select a message to be attached to the rule. For details refer to the Configurator Messages section.

Configurator Messages

You use this option to set up and amend the messages you use with your rules.

You can attach specific warning or error messages to Configurator rules via the Configurator Rules option. You can also display messages on any violation of maximum or minimum quantities for components on a configuration (via the Route / Structure option within Production).



Note: If specific messages are not specified, standard messages are displayed for rule or component quantity errors during configuration.

Each message must be assigned to a message category. These categories must have been set up in Inventory Descriptions (MCAT) already. You can use the categories to help you search for a particular message. You can if you wish define and use one category only. The categories must be codes of not more than 5 characters in length.

Component Costing



Warning: Not used for pricing methods 0 and 1.

If you are using pricing method 2 (cost plus), then you will want to transfer your item costs to a price list (to be called a 'cost list' to distinguish it from normal price lists) so they can be picked up when pricing.

This option writes these costs to a price list automatically. Note that the price profile must first have been set up in Sales Order Processing, and you must be authorised to the profile before you can use it.

All costs are in base currency.

Configurator Component Batch Costing Screen



Select Component Costing to display the Configurator Component Batch Costing screen.

You use this screen to enter the costing details for a specified batch.

CN300	Z1 - Demonstration Company	User V3USER	17/10/95
			15:45:23
Configurator Component Batch Costing - Select			
Select a range of Model Items or leave blank for all			
Model Item...?... to			
Specify the following:			
Price list.....?... ..			
Effective date..... 17/10/95			
F3=Exit F4=Prompt			
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Fields:

Model Item

Enter the range of model items for which you wish to create cost price lists.

Price List

Enter the list to be created.

Effectivity Date

Enter the date from which the cost price list is to be used.

Once submitted, the batch job creates cost price list details.

Route Maintenance

The route / structure maintenance function in the Production Definition Management module of Production is used by Configurator to define the operations, components (generic and actual) and text for Configurator model items and order configured items which will appear as user choices during the process of configuration.

Extra options specific to Configurator are presented while you are entering your route, as long as all of the following conditions are satisfied:

- Configurator is attached for the company
- The item being maintained is a model item or configured item
- The route code is the configuration route for the company



Note: For details of the definition of model items, configured items and the configuration route, refer to the Configurator Company Profile Maintenance section.

For details of the standard options available in Route Maintenance, refer to the Production Definition Management product information.

Each route can contain:-

- Standard and optional operations
- Standard and optional components
- Generic component (e.g. 'paint'. An actual component for the generic 'paint' could be 'green matt paint'), either mandatory or optional
- Rules to be applied to the product
- Default option for features
- Minimum and maximum quantities of a component
- Presentation sequence of operations, components and features
- Operations and features are specified as being either preconfigured or selectable when the order is placed.

Route Maintenance Screen



Select Route / Structure to display the selection screen.

Enter your model item and the configuration route, and press **Enter** for the header. Complete the header screen as necessary (refer to the PDM product information for more details), and press **Enter** again to see the Route Maintenance Details screen, which will not have route details showing if you entered a new model or a new order configured item.

DB132		Z1 - Demonstration Company		USER V3USER		17/10/95 15:47:25	
Route Maintenance							
Parent: DESK				Std Lot Size		1.000	
Route: C				Receive into Stockroom 01		UOM EA	
0 Opsq	Insq	Input	ST UM	Description	Quantity	Shr%	
-	10			ASSEMBLE DESK		.0	
-	10	10 GNSIDE	01	EA Deskside - generic	2.000	.0	
-	10	20 GNTOP	01	EA Desktop - generic	1.000	.0	
-	10	50 SCREW01	01	EA Screw 01	4.000	.0	
-	20			PAINT ASSEMBLY		.0	
-	20	10 GNPNT01	01	EA Paint 01 (Generic)	1.000	.0	
-	30			FIX COMPUTER SHELF		.0	
-	30	10 COMSHEL	01	EA Computer shelf	1.000	.0	
1=Maintain 2=Add Input Item							
Enter Sequence to maintain or Position to:							
F3=Exit F8=Replace route F12=Cancel F17=Copy route F19=Re-sequence							
F21=Presentation Sequence							
20-30 SF MW KS IM II KENB20A PET147S1							

Select an operation with 1 to see the operation details screen. This screen has only one Configurator field:

Fields:

Configurator Op Id

Enter here the unique code for this operation which will identify it to any rules.

Every operation within a configuration route must have its own operation ID set up under the major type 'CNOI' in Inventory Descriptions this is so that any rules know to which operation they are attached.



F22 enables you to maintain formatted text



Enter the operation details and press **Enter**, to display the Configuration Options window.

Configuration Options Window



Enter the operation details and press **Enter**, to display the Configuration Options window.

This window allows you to change the Configuration Options.

```

DB134                Z1 - Demonstration Company        USER V3USER        17/10/95
                                                16:15:16

Route Operation Maintenance        UPDATE

Parent..... DESK
Operation Sequence.. 10
Configurator Op Id.?. DES
Op. Description..... ASS
Work Station.....?.. WSO
Work Centre.....?..
Duration Basis.....?..
Time Basis Code....?..
Move Days.....
Transfer Batch Size...
SET-UP DETAILS Labour.... .0000 Mins      Crew Size.. 1.00
RUN DETAILS Machine.. .0000 Mins/lot
Labour.. .0000 Mins/lot      Crew Size.. 1.00

Sub-Contract..... (1=Yes)
Sub-Contract Cost. .00000 Shrinkage %.. .0

* This Operation has inputs associated with it.
F4=Prompt F11=Delete this line F12=Replace line without update.
F21=Operation Text F22=Formatted Text
00-55  S2 MW KS IM II KENB20A PET147S1
  
```

Fields:

Type

Enter one of the following:

0 - Operation is optional.

1 - Operation is mandatory. Mandatory operations must be part of a configuration and cannot be deleted.

Presentation Sequence

Specify the sequence in which the operation is to appear during configuration.

This defaults to the normal operation sequence.

Rule Section

Enter the relevant rule section.

Rule section is used to group together sets of Configurator rules so that they may be applied to a particular section of the route. This field defaults to the operation sequence number.

Order Entry Option

Enter one of the following:

0 - the operation is a preconfigure option.

1 - the operation is an order entry option.

Specify whether the operation is selectable in preconfiguration mode or in order entry mode. This option still has relevance even if you entered 1 in the Type field above to indicate a mandatory operation.

If an order configured item is being defined, this field is fixed at 1, because no preconfiguration is to take place.



Note: The maximum and minimum component options are for memo purposes only and are not currently used elsewhere in the system.



Press Enter to save the details to a work file. To update the details press F8 on the Operation / Component selection screen.

Functions:

F20

Specify Configurator rules to apply to this section of the route.

Input Maintenance Window



Select a component on the route screen and press **Enter** to display the Input Maintenance window.

If Configurator is attached and a model or configured item is being defined to the configuration route, extra Configurator options will be available. Configurator options defined to generic components will apply to the actual components selected to replace those generics during configuration. F20 is available to define Configurator rules to the route.

DB132

Z1 - Demonstration Company

USER V3USER

17/10/95

Route Maintenance

Parent: DESK

Route: C

0 Opsq Insq Input

10

1 10 10 GNSIDE 01 E

10 20 GNTOP 01 E

10 50 SCREW01 01 E

20

20 10 GNPNT01 01 E

30

30 10 COMSHEL 01 E

Receive

ST U

1=Maintain 2=Add Input Item

Enter Sequence to maintain

F3=Exit F8=Replace route F12

F21=Presentation Sequence

06-61

SF

MW

Input Maintenance

Operation 10 Input Seq. 10 UPDATE

Input..... GNSIDE

Issuing Stockroom...?.. 01

Quantity Per Lot..... 2.000

Fixed Quantity Per..... (1=Yes)

Shrinkage %..... .0

Change Reference.....

Effective From Date....

To Date..... 999999

Type..... 1 (0=Opt,1=Mnd)

Allow Qty Override..... 0 (0=No,1=Yes)

Presentation Sequence.. 10

Rule Section..... 0010

Order Entry Option..... 0 (0=No,1=Yes)

Min Qty.. .000 Msg?

Max Qty.. .000 Msg?

Default Input.....?..

F4=Prompt F11=Delete F12=Cancel

F20=Rules

KS

IM

II

KENB20A

PET147S1

Fields:

Route Operation Component

Enter a component item that is required at this Production stage. All components that are required at this stage of the process should be defined within this operation. In this way, whenever a particular operation is due to begin, the system can ensure that all the necessary components are available when they are really required, thus avoiding excessive stocks of components being held before they need to be consumed.

Issuing Stockroom

Enter the stockroom from which this component will be issued to this operation.

If left blank, the system will use the primary stockroom as defined on the item master record.

All material requirements planning using this route will place a requirement on this stockroom for the component.

Component Quantity Per Parent Lot

Enter the standard quantity of this component that is required by this operation to produce the standard parent lot size.

Fixed Quantity Per

1 - Quantity is fixed.

Blank - Quantity required is dependent on the lot size of the parent.

Component Shrinkage Percentage

Enter the percentage which is expected to be lost or wasted as a result of this process. The system will use this figure to extend the quantity per when calculating the quantity required for the operation. It is also used as part of the standard cost roll up to establish the correct amount of component usage in an item.

See the Production product information for extended details of shrinkage and examples of material requirement calculations.

Change Reference

Enter a reference to group together route / operation / component combinations (e.g. as a result of an engineering change which affects a range of items or routes). The field can hold an Engineering Change Note or a Drawing reference.

Effective From Date

Enter the first date on which this component is effective within this operation. If the field is left blank it will be assumed that there is no start date limitation on the use of this component.

Effective To Date

Enter the date on which this component ceases to be effective in this operation. If left blank, the component is assigned an open-ended expiry date.



All the following fields (except the final component sequence field) are specific to Configurator:

Configuration Type

0 - Component is optional.

1 - Component is mandatory.

Specify whether the component is mandatory or optional within the configuration. Mandatory components must be part of a configuration if the operation to which they are defined is itself included in the configuration.

Generic items may well be defined as mandatory. This does not mean that the generic item itself must be included, but that one of its actuals must be.

Allow Quantity Override

0 - The quantity per is fixed in Configurator.

1 - The quantity per is variable.

Specify whether the 'quantity per' for the component can be maintained during configuration.

Presentation Sequence

Specify the sequence in which the component is to appear within its operation during configuration.

This setting can be crucial as it may control the order in which rules are presented to the operator. This affects the actual logic of the 'program' which the rules constitute.

Refer to the section dealing with rules maintenance for further details regarding this issue.

The field defaults to the Production sequence.

Rule Section

Rule section is used to group together sets of Configurator rules so that they may be applied to a particular section of the route. Rule section defaults to the operation sequence number.

Order Entry Option

Enter one of the following:

0 - The component is selected in preconfiguration.

1 - The component is selected in order entry.

For order configured items, this is fixed at 1.

Minimum Quantity

If quantity override is allowed for the component, specify the minimum permitted quantity per one of the parent item.

Minimum Quantity Message

Optionally specify the code of a Configurator message which is to display if a quantity per less than the permitted minimum is entered for the component during configuration. If no specific message is entered a standard message will display. A prompt is available on valid Configurator messages which also allows the option to create a new message.

Maximum Quantity

If quantity override is allowed for the component, specify the maximum permitted quantity per one of the parent item.

Maximum Quantity Message

The function works as for minimum quantity, if the quantity is greater than the permitted maximum.

Default Component

If the component being maintained is a generic component, a default actual component may optionally be specified to replace the generic component during configuration. The default component and generic component must be in the same search family.

The default component may be replaced during configuration by any other actual component within the same search family. A prompt is available on items defined to Production.

Component Sequence Increment

When adding multiple component records the increment between component sequence numbers may be set. The default is 10 if the starting sequence is a multiple of 10, otherwise the default is 1.



Press Enter to write the details to a work file. To update the details then press F8 on the Operation / Component selection screen.

Functions:

F11

Delete the component from this operation on the route.

F20

Define Configurator rules to apply to this section of the route.

Apply Rules to Route Window



To apply rules to either operations or components in the route, press F20 from the window of Configurator operation details or the component entry window.

This window displays existing Configurator rules applied to this section of the route and allows further rules to be applied. A rule may also be selected for removal from the route or for enquiry on rule details.

DB132

Z1 - Demonstration Company

USER V3USER

17/10/95

Route Maintenance

Parent: DESK

Route: C

0 Opsq Insq Input

10

1 10 10 GNSIDE

10 20 GNTOP

10 50 SCREW01

20

20 10 GNPNT01

30

30 10 COMSHEL

1=Maintain 2=Add Input

Enter Sequence to ma

F3=Exit F8=Replace route

F21=Presentation Sequence

09-27

SF

MW

KS

IM

II

KENB20A

PET147S1

Input Maintenance

Operation 10 Input Seq. 10

UPDATE

Receive

Input..... GNSIDE

Apply Rules to Route

Rule Test Section: 0010

Seq	Rule	Description
1	DSK015	Comp shelf needs adapted top
2	DSK020	Desk all PVC or all pine
3	DSK025	Desk all PVC or all pine

4=Delete 5=Enquire

F4=Prompt F12=Previous

F4=Prompt F11=Delete F12=Cancel

F20=Rules

Fields:

Rule Test Section

The section of the route for which rule tests will be applied for the rules displayed in the list. Rule section is one of the Configurator options maintained against operations and components on the route.

Sequence

The sequence number of the rule in the list. This affects the order in which rules will be actioned for the route.

Rule Code And Description

The code of the Configurator rule and the description associated with it in Rules Maintenance.

Option Select

Enter 4 to remove a rule from the route. Enter 5 to enquire on rule details.



Note that option 4 does not delete the definition of a rule, but removes it from this particular route only.

Sequence Number For Addition

When adding a new rule to the route, specify the sequence in which it is to appear in the list and be processed during configuration.

Rule Code For Addition

When adding a new rule to the route, specify the code associated with the rule. A prompt is available on valid Configurator rules.

Select Format Window

- To maintain the Configurator formatted text press **F22** from the route operation details screen. In the window displayed you can select and sequence one or more text formats for the operation.

You use this window to select the text format you want to maintain.

08134

Z1 - Demonstration Company

USER V3USER

24/10/95
15:44:16

Route Operation Maintenance

UPDATE

Select Text Format

Position to...

1 Seq Fmt Description

Con 1 CHK PRODUCT CHECKLIST

Op. OP1 MINOR : Operation

Wor

Wor

Dur

Tim

Mov

Tra

F8=Update F12=Previous

More:

me

ty... (Basis 6)

/Est... 1 (1=Studied 2=Est)

Time.. .0000 Minutes

Crew Size.. 1.00

Crew Size.. 1.00

SET-UP DETAILS Labour.... .0000 Mins

RUN DETAILS Machine.. .0000 Mins/lot

Labour.. .0000 Mins/lot

Sub-Contract..... _ (1=Yes)

Sub-Contract Cost. .00000 Shrinkage %.. .0

* This Operation has inputs associated with it.

F4=Prompt F11=Delete this line F12=Replace line without update.

F21=Operation Text F22=Formatted Text

01-02 **SF** MW KS IM II KENB20A PET149S1

Fields:

Format Code And Description

The code of the text format and its associated description as defined in Production text maintenance.

Position To

To position the list to a particular text format, specify a text format code and press Enter.

Select

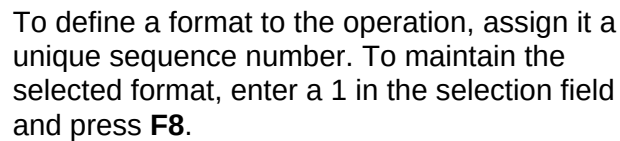
Enter 1 to select a text format from the list.

Sequence

Enter a sequence number to define the order in which the selected text format is to appear during configuration.

- To define a format to the operation, assign it a unique sequence number. To maintain the selected format, enter a 1 in the selection field and press **F8**. When maintenance is complete, press **F12** to return to the route operations detail

screen.



Heading

Displays or allows maintenance of a heading for the text line, depending on the option specified for the format.

Text

Enter up to the permitted number of lines of text for this format.

Position To Line Number

Specify a line number at which to position the text.

Insert Lines

Specify the number of new lines to be inserted.

Insert After Line Number

Specify the line number after which to insert new lines.

Production Text Types

The Production Definition Management module of Production enables text to be maintained against routes, operations, work stations, cost centres, work centres and production orders.

If Configurator is attached for the company, an extra facility is available to maintain formatted text for operations on the configured route. Each format corresponds to a text sub-type within the operation text type defined for the company.

This section refers only to the definition of operation text formats for Configurator. For full details of text management within the Production systems, refer to the Production Definition Management product information.

To maintain a Configurator formatted text format, enter the text major type which corresponds to operation text for this company, together with an existing or new sub-type. Each sub-type defines a particular format for the operation text.

A prompt is available on valid text types.



When you enter valid details and press Enter, a screen displays which you can use to maintain the text sub-type details.

If the sub-type is not known, enter the major type and press F17 to display all sub-types for the major type.

Functions:

F17

Display a list of sub-types defined to the entered major type.

Text Type Maintenance Screen 1

This screen displays a list of sub-types currently set up for the major type and allows the selection of individual sub-types for maintenance.

TX001		Z1 - Demonstration Company		USER V3USER		17/10/95 16:47:12	
Text Type Maintenance							
Subtype		Description		Major type		OPER - MAJOR : Operation	
CHK					No. Lines	Text Length	O/E Optn
OP1	MINOR : Operation				0000	00	
Enter Sub-type To maintain or add ☐ CHK							
F4=Prompt F12=Re-select major type							
21-40 ☐ MW KS IM II KENB20A PET147S1							

If Configurator is attached and operation text is being maintained, each sub-type represents an individual format which can be attached to an operation on the configured route in Production route maintenance. The format options then displayed against each format are as follows:

Fields:

Number Of Lines

The maximum number of lines of text allowed for the format.

Text Length

The maximum number of characters of text allowed per line.

O/E Option

0 - a preconfiguration option

1 - an order entry option

Enter the mode of configuration in which text for the format is maintainable.



To maintain an existing or new format, enter the required sub-type code and press **Enter**.

Text Type Maintenance Screen 2

This window allows options for Configurator formatted operation text to be specified.

TX001	Z1 - Demonstration Company	USER V3USER	17/10/95
			16:47:12
Text Type Maintenance			
		Major type	OPER - MAJOR : Operation
Subtype	Description	No. Lines	Text Length
CHK		0000	00
OP1	MINOR : Operation	0000	00

Sub-type Description.

*CHANGE

PRODUCT CHECKLIST

Number of Lines..... 4

Text Length..... 60

Heading Override..... 0

Order Entry Option... 1

Print on Works Order. 1

Quantity Sensitive... 0

F11=Delete Sub-type F12=Previous

Enter Sub-type To maintain or add C

F4=Prompt F12=Re-select major type

20-67 SF MW KS IM II KENB20A PET147S1

The following details may be maintained:

Fields:

Sub-Type Description

Enter or amend a description for this category of text.

Number Of Lines

Specify the number of lines of text allowed when this format is selected for the entry of formatted text.

Text Length

Specify the number of characters allowed on each line when this format is selected for the entry of formatted text.

Heading Override

0 - Heading text is only maintainable in route operation maintenance

1 - Heading text is also maintainable during configuration.

Specify whether heading text for each line of formatted text is maintainable during configuration.

Order Entry Option

0 - The format is a preconfiguration option.

1 - The format is an order entry option.

Specify whether this format is an order entry or a preconfiguration option when selected for the entry of formatted text.

Print On Production Order

0 - Do not print text.

1 - Print text.

Specify whether text for this format is to be printed on production order documentation when entered for a configured order line.

Quantity Sensitive

This field is not currently used in the standard system.



When you enter valid details and press Enter,
the format details are updated.